Practice Focus: Enterprise ERP Architecture & Industrial Telemetry

PRIXGEN INDUSTRIAL RESEARCH | EXECUTIVE WHITE PAPER

[*] Two Silent Margin Leaks: Shooting purge waste math and ultrasonic wall-thickness paradox calculation

FROM POLYMER TO PIPE:

Building a Connected Operating Model

For PVC and Plastics Pipe Manufacturers

A comprehensive framework for eliminating operational disconnects,

[*] The 10-Step Extrusion Control Journey: Direct telemetry control points from bulk silo to yard bundle

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Published by Prixgen Industrial Intelligence | prixgen.com

EXECUTIVE SUMMARY CORE TAKEAWAYS:

[*] 6 Disconnected Systems: Unifying Procurement, Formulation, Production, Quality, Inventory & Finance

continuous value stream. The true cost surfaces in three ways:

1. One Value Chain, Six Disconnected Systems

procurement directly to machine telemetry, quality gates, and dispatch invoicing."

For most PVC and plastics manufacturers, operations are managed across six isolated systems,

1. Procurement Silo: Resin price volatility tracked in isolated supplier spreadsheets.

It is connected decision-making across the entire value chain linking raw resin

The Cost of Operational Disconnect

When these six functions operate in silos, the manufacturer loses the ability to see itself as a single

THE STRATEGIC PRINCIPLE:

"The future of plastics manufacturing is not automated production alone.

2. The 10-Step Extrusion Control Journey

Every finished pipe passes through ten distinct stages. A connected ERP captures digital telemetry

Stage

01

Prixgen's Odoo ERP architecture creates an unbroken audit trail linking ten critical nodes:

6. Shift Operator --> 7. QC Burst Record --> 8. Finished Bundle --> 9. Yard Bin --> 10. Dispatch Note

Defensible 10-Link Batch Traceability Chain

A single defective pipe can trigger catastrophic liability unless the origin is defensibly proven.

1. Supplier COA --> 2. Silo Lot --> 3. Compounding Batch --> 4. Work Order --> 5. Extruder Line

Result: Defect root cause isolated in under 3 minutes without recalling unaffected production batches.

3. Two Silent Margin Leaks in Plastics Manufacturing

Leak #1: The Wall-Thickness Paradox (Resin Over-Give)

In PVC pipe production, standards define a MINIMUM wall thickness. Operators intentionally

and stabilizer dosing ratio in compounding batches based on exact regrind availability.

The Odoo Solution: Real-Time Ultrasonic Telemetry & Weight-Per-Meter Feedback

Continuous weight-per-meter sensors automatically adjust haul-off traction speed to maintain

Leak #2: Unmodelled Shooting Purge Waste & Regrind Drift

Die head changeovers, color switches, and cold starts generate kilograms of degraded polymer purge.

The Odoo Solution: Closed-Loop Scrap Reclaim & Auto-Balanced MRP BOMs

Purge weight is logged at extruder bins via barcode. Odoo MRP dynamically adjusts the virgin resin

4. The 5-Layer Connected Operating Model

Prixgen deploys a unified 5-layer architecture to bridge shop-floor hardware with executive decisions:

LAYER 5: Dealer & Commercial Portal

Live secondary sales, automated volume rebates, claims validation & digital dispatch notes.

Primary Metric

Resin Price Variance

Real-Time Manufacturing Scorecard

Management steering requires live transaction metrics rather than stale monthly spreadsheets:

Operational Domain

Procurement

5. Next Steps: Schedule a Plant Diagnostic Audit

Transforming a PVC or plastics plant into a connected manufacturing operation starts with

Phase 1: Plant & Silo Diagnostic (Week 1-2)

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Audit raw material weignbridge gaps, formulation leakages, and yard inventory accuracy.

Ready to Connect the Journey From Polymer to Pipe?

Speak with our manufacturing architects about implementing this framework in your plant.

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Document ID: PX-WP-PVC-2026-09 | Classification: Public Executive Distribution